

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination - Winter 2025

Course: B. Tech

Branch: Common To All Branches

Semester: I

Subject Code & Name: BTES103 & Engineering Mechanics

Max Marks: 60

Date: 22/01/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

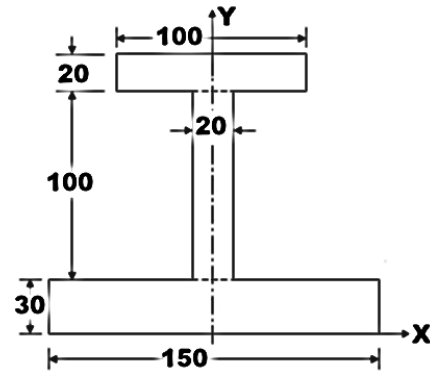
	(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 A rigid body is acted upon by a force system whose resultant force is zero but resultant moment is non-zero. The body will experience:	Understand / CO2	1
A) Pure translation		
B) Pure rotation		
C) Translation and rotation		
D) Complete equilibrium		
2 The moment of a force about a point depends on:	Remember / CO2	1
A) Magnitude of force only		
B) Direction of force only		
C) Perpendicular distance from point to line of action		
D) Area of cross-section		
3 The main purpose of idealizing an engineering problem is to:	Understand / CO2	1
A) Increase complexity		
B) Make mathematical analysis easier		
C) Avoid assumptions		
D) Eliminate forces		
4 A coplanar force system is in equilibrium if:	Understand / CO2	1
A) $\sum F_x = 0$ only		
B) $\sum F_y = 0$ only		
C) $\sum F_x = 0$, $\sum F_y = 0$, and $\sum M = 0$		
D) Resultant force is minimum		
5 The centroid of a composite section lies outside the material area when:	Understand / CO2	1
A) Section is symmetric		
B) Material density varies		
C) Loads are eccentric		
D) The section contains voids or cut-outs		
6 Friction always acts in a direction:	Remember / CO2	1
A) Opposite to relative motion or tendency of motion		
B) Perpendicular to motion		
C) Same as applied force		
D) Vertically downward		

7	In a plane truss, if the number of members m and joints j satisfy $m < 2j - 3$, the truss is:	Understand / CO2	1
	A) Perfect		
	B) Redundant		
	C) Indeterminate		
	D) Deficient and unstable		
8	The method of sections is more efficient than method of joints when:	Understand / CO2	1
	A) Truss has many members		
	B) Reactions are unknown		
	C) Forces in only selected members are required		
	D) Truss is statically indeterminate		
9	In projectile motion, the horizontal component of velocity remains constant because:	Understand / CO2	1
	A) Air resistance is absent		
	B) No acceleration acts horizontally		
	C) Gravity acts vertically downward		
	D) Both B and C		
10	If two particles move with equal velocities in the same direction, their relative velocity is:	Remember / CO2	1
	A) Zero		
	B) Maximum		
	C) Equal to twice the velocity		
	D) Dependent on acceleration		
11	D'Alembert's principle introduces an inertia force which is:	Understand / CO2	1
	A) Equal and opposite to applied force		
	B) Equal to mass $\times$ acceleration and opposite in direction		
	C) Acting in the direction of acceleration		
	D) Equal to momentum change		
12	In an elastic collision between two bodies:	Remember / CO2	1
	A) Momentum is conserved but kinetic energy is not		
	B) Kinetic energy is conserved but momentum is not		
	C) Both momentum and kinetic energy are conserved		
	D) Neither momentum nor energy is conserved		
Q.2 Solve the following.			12
A)	Discuss idealisations are made in engineering mechanics.	Understand / CO1	6
B)	Explain types of supports and loads with neat free body diagrams.	Understand / CO1	6

Q.3 Solve the following.

12

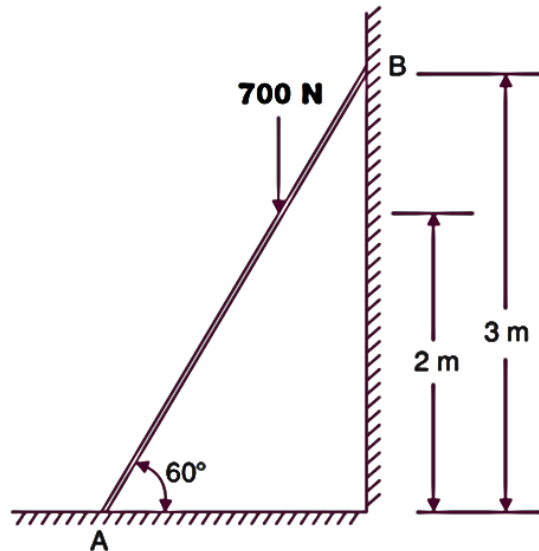
- A) Locate the centroid of the I-section shown in figure with respect to the axes shown. (All dimensions are in mm)



Apply /
CO3

6

- B) A ladder weighing 100 N is to be kept in the position shown in figure, resting on a smooth floor and leaning on a smooth wall, also a man weighing 700 N is at 2m above floor level. Determine (i) The horizontal force F required at floor level to prevent it from slipping. (ii) If the horizontal force F is to be applied at a height of 1 m above the ground level, how much should F be?



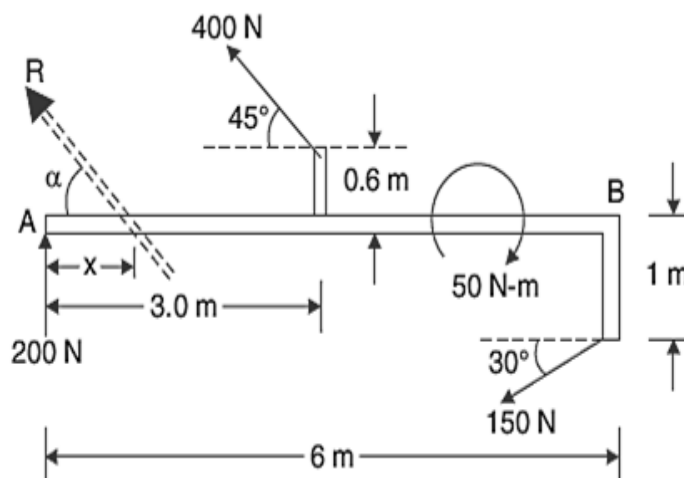
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Q.4 Solve Any Two of the following.

12

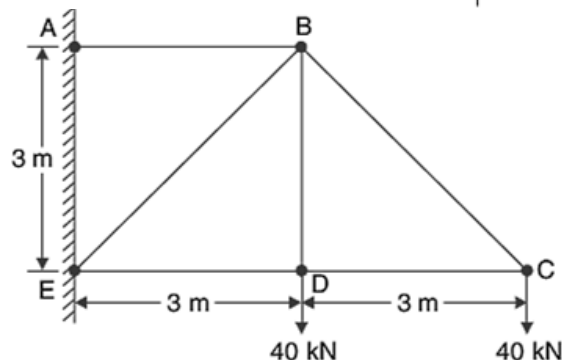
- A) A bracket is subjected to three forces and a couple as shown in figure. Determine magnitude, direction and the line of action of the resultant.



Apply /
CO2

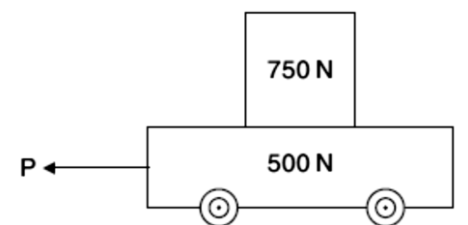
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- B) Find the forces in all the members of the truss shown in figure & tabulate the results.



Apply /
CO2

6

C)	Explain classification of trusses and assumptions made in truss analysis. Compare the method of joints and method of sections.	Understand / CO2	6
Q.5 Solve Any Two of the following.			12
A)	A particle is projected vertically upwards from the ground with an initial velocity of u m/sec. Find (i) the time taken to reach the maximum height; (ii) the maximum height reached; (iii) time required for descending; and (iv) velocity when it strikes the ground.	Apply / CO4	6
B)	A rocket is released from a jet fighter flying horizontally at 1200 kmph at an altitude of 3000 m above its target. The rocket thrust gives it a constant horizontal acceleration of 6 m/s^2 . At what angle below the horizontal should pilot see the target at the instant of releasing the rocket in order to score a hit?	Apply / CO4	6
C)	A passenger train is 240 m long and is moving with a constant velocity of 72 kmph. At a particular time, its engine approaches last compartment of a goods train moving on a parallel track in the same direction. 25 seconds later its engine starts overtaking the engine of goods train. It took 30 seconds more to completely overtake the goods train. Determine the length and speed of the goods train.	Apply / CO4	6
Q. 6 Solve Any Two of the following.			12
A)	A 750 N crate rests on a 500 N cart. The coefficient of friction between the crate and the cart is 0.3 and between cart and the road is 0.2. If the cart is to be pulled by a force P such that the crate does not slip. Using D' Alembert's principle, determine: (i) the maximum allowable magnitude of P , (ii) the corresponding acceleration of the cart.	Apply / CO5	6
			
B)	A pump lifts 40 m^3 of water to a height of 50 m and delivers it with a velocity of 5 m/sec. What is the amount of energy spent during this process? If the job is done in half an hour, what is the input power of the pump which has an overall efficiency of 70%?	Apply / CO5	6
C)	Direct central impact occurs between a 300 N body moving to the right with a velocity of 6 m/sec and 150 N body moving to the left with a velocity of 10 m/sec. Find the velocity of each body after impact if the coefficient of restitution is 0.8.	Apply / CO5	6

*** End ***